# A hitch-hiker's guide to the metaplectic trace formula for $SL(2)$

Sections 2–3, revised for the double cover (draft)

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These are the revised Sections 2 and 3 of the 1995 notes, restricted to $n = 2$, compiled on their own. In Section 2, Lemmas 2.1–2.9 keep the numbers of the 1995 notes and Lemma 2.10 is new. Section 3 is shorter than in 1995, because for $n = 2$ the cover of a torus is abelian; the old and new numbers are matched in `revision_log.md`, which also lists the changes.

## 2 Background on the metaplectic double cover

This section fixes the double cover of $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$ and the conventions used in the rest of the paper. Section 2.1 treats a single place: Kubota's cocycle, the splittings over unipotent and compact subgroups, the center, and the preimages of tori. These are then assembled into the cover of $G(\mathbb{Q}_p)$. Section 2.2 builds the adelic cover and its splitting over $G(\mathbb{Q})$, and Section 2.3 fixes a double cover of $GL(2)$ and the resulting $GL(2)$-action on the cover of $SL(2)$, which is what Section 7 uses.

From now on $n = 2$. The 1995 notes treated the $n$-fold cover; for $n = 2$ several statements become simpler, and one, the commutativity of preimages of abelian subgroups (Lemma 2.2(g)), becomes a two-line argument. What changes for general $n$ is listed in Remark 2026.4.

### Notation

Let $k$ be a number field. (For $n = 2$ the hypothesis $\mu_n \subset k$ of the 1995 notes always holds, and $k$ may have real places.) Let $\Sigma_k$ be the set of places of $k$, $\Sigma_k^\infty$ the archimedean places and $\Sigma_k^f$ the finite places. Let $\mathbb{A}$ be the adele ring of $\mathbb{Q}$ and $\mathbb{A}_k$ that of $k$. For $v \in \Sigma_k$ let $k_v$ be the completion. For $F = k_v$ non-archimedean let $\mathcal{O} = \mathcal{O}_F$ be its ring of integers, $\varpi = \varpi_F$ a uniformizer, $q$ the order of the residue field and $\mathrm{ord} = \mathrm{ord}_F \colon F^\times \to \mathbb{Z}$ the normalized valuation, so $\mathrm{ord}(\varpi) = 1$ and $|\varpi| = q^{-1}$. We call $v$ *dyadic* if $v \mid 2$, and put $e_v = \mathrm{ord}_v(2)$. Absolute values are normalized; at a complex place $|z|_v = z\bar{z}$.

Let $(a, b) = (a, b)_v$ be the quadratic Hilbert symbol of $F = k_v$ [W1, Se]. We use only the following properties:

(H1) $(\cdot, \cdot)$ is bimultiplicative and symmetric, with values $\pm 1$; hence $(a^{-1}, b) = (a, b)$.

(H2) $(a, b) = 1$ if $a$ or $b$ lies in $F^{\times 2}$; $(a, -a) = 1$; and $(a, 1 - a) = 1$ for $a \neq 0, 1$ (the Steinberg relation).

(H3) If $v$ is finite and not dyadic and $a, b \in \mathcal{O}^\times$, then $(a, b) = 1$. If $v$ is complex, $(\cdot, \cdot) \equiv 1$. If $v$ is real, $(a, b) = -1$ exactly when $a < 0$ and $b < 0$.

(H4) (Product formula) For $a, b \in k^\times$, $(a,b)_v = 1$ for almost all $v$ and $\prod_v (a,b)_v = 1$.

By (H3), $(-1,-1)_v = 1$ at non-dyadic finite places and at complex places, and $(-1,-1)_v = -1$ at real places. At a dyadic place both values occur; $(-1,-1)_{\mathbb{Q}_2} = -1$.

Let $G$ be the group over $\mathbb{Q}$ obtained from $SL(2)$ by restriction of scalars from $k$ to $\mathbb{Q}$, so $G(\mathbb{Q}) = SL(2,k)$ and $G(\mathbb{Q}_p) = \prod_{v|p} SL(2,k_v)$.

## 2.1 Local aspects

Here $F = k_v$ for one place $v$. We define the cover, compute it on the subgroups that matter later (unipotent, compact, diagonal, central, anisotropic), and then pass to $G(\mathbb{Q}_p)$.

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,F)$ put $x(g) = c$ if $c \neq 0$ and $x(g) = d$ if $c = 0$, and define $\beta = \beta_v \colon SL(2,F) \times SL(2,F) \to \mu_2$ by

$$\beta(g_1,g_2) = \Big( \frac{x(g_1g_2)}{x(g_1)}, \frac{x(g_1g_2)}{x(g_2)} \Big) = \big( x(g_1), x(g_2) \big) \big( -x(g_1)^{-1} x(g_2),\, x(g_1g_2) \big). \tag{2.1}$$

At a complex place $\beta \equiv 1$ by (H3).

**Lemma 2.1** (Kubota [K1], Moore [M]). *$\beta$ is a continuous Borel 2-cocycle, $\mu_2$ carrying the discrete topology. Its class in $H^2(SL(2,F), \mu_2)$ defines a topological central extension*

$$1 \to \mu_2 \to \overline{SL(2,F)} \xrightarrow{p} SL(2,F) \to 1,$$

*which is nontrivial unless $F = \mathbb{C}$.*

The two expressions in (2.1) agree, and the cocycle identity holds; both are checked on random elements over $\mathbb{R}$, $\mathbb{Q}_2$, $\mathbb{Q}_3$, $\mathbb{Q}_5$ and $\mathbb{Q}_7$ in `checks/sec2_n2_check.py`. For $F = \mathbb{R}$ the group $\overline{SL(2,\mathbb{R})}$ is the metaplectic group $Mp(2,\mathbb{R})$.

For a subgroup $H \subset SL(2,F)$ put $\overline{H} = p^{-1}(H)$. Elements of $\overline{SL(2,F)}$ are written $(g,\zeta)$ with $g \in SL(2,F)$, $\zeta \in \mu_2$, and multiplied by

$$(g_1,\zeta_1)(g_2,\zeta_2) = (g_1g_2,\ \beta(g_1,g_2)\zeta_1\zeta_2). \tag{2.2}$$

Since $\zeta^{-1} = \zeta$, inverses and conjugates are

$$(g,\zeta)^{-1} = (g^{-1}, \beta(g,g^{-1})\zeta), \qquad (g,\zeta)(h,\zeta')(g,\zeta)^{-1} = \big( ghg^{-1},\ \beta(g,h)\beta(gh,g^{-1})\beta(g,g^{-1})\zeta' \big). \tag{2.3}$$

Let $B \subset SL(2)$ be the upper triangular Borel subgroup, $N$ its unipotent radical and $A$ the diagonal torus. Write $a(t) = \mathrm{diag}(t,t^{-1})$ and $n(u) = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$. For $\theta \in F^\times \smallsetminus F^{\times 2}$ let

$$T_\theta = \left\{ \begin{pmatrix} a & b\theta \\ b & a \end{pmatrix} \;\middle|\; a,b \in F,\ a^2 - \theta b^2 = 1 \right\} \cong E^1, \qquad E = F(\sqrt{\theta}),$$

the norm-one group of $E$. Every anisotropic maximal torus of $SL(2,F)$ is $GL(2,F)$-conjugate to some $T_\theta$.

If $v$ is finite let $K = K_0 = SL(2,\mathcal{O})$ and, for $m \geq 1$,

$$K_m = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathcal{O}) \;\middle|\; a-1,\ d-1,\ b,\ c \in \varpi^m \mathcal{O} \right\}.$$

If $v$ is non-dyadic, define $\kappa = \kappa_v \colon K \to \mu_2$ by

$$\kappa\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} (c,d)_v, & 0 < |c|_v < 1, \\ 1, & \text{otherwise.} \end{cases} \tag{2.4}$$

If $v$ is archimedean let $K = SO(2)$ ($v$ real) or $K = SU(2)$ ($v$ complex).

**Lemma 2.2.** *Let $F = k_v$.*

*(a) $\beta \equiv 1$ on $N(F) \times N(F)$, so $n \mapsto (n,1)$ is a homomorphism and $\overline{N(F)} = N(F) \times \mu_2$.*

*(b) Let $v$ be finite and non-dyadic. Then $\beta(k_1,k_2) = \kappa(k_1)\kappa(k_2)\kappa(k_1k_2)$ for $k_1, k_2 \in K$, so $k \mapsto (k, \kappa(k))$ is a homomorphism and $\overline{K} = K^* \times \mu_2$ with $K^* = \{(k,\kappa(k))\}$. The restriction of $\beta$ to $K \times K$ is not identically 1, so the set $\{(k,1) \mid k \in K\}$ is not a subgroup.*

*(c) Let $v$ be finite and $m \geq 1$ with $1 + \varpi^m \mathcal{O} \subset F^{\times 2}$. Then $\beta \equiv 1$ on $K_m \times K_m$, so $k \mapsto (k,1)$ splits $\overline{K_m}$. The hypothesis holds for every $m \geq 1$ if $v$ is non-dyadic, and for $m \geq 2e_v + 1$ if $v$ is dyadic.*

*(d) Let $v$ be archimedean. If $v$ is complex, $\overline{SL(2,F)} = SL(2,F) \times \mu_2$. If $v$ is real, $\beta \equiv 1$ on $A(\mathbb{R})^0 \times A(\mathbb{R})^0$, so $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$, while $\overline{SO(2)}$ is connected, the nontrivial double cover of the circle [Ge2].*

*(e) The center of $\overline{SL(2,F)}$ is $\overline{\{\pm 1\}}$, a group of order 4 in which $(-1,\zeta)^2 = (1,(-1,-1)_v)$. It is cyclic when $(-1,-1)_v = -1$ (for example $F = \mathbb{R}$ or $F = \mathbb{Q}_2$) and is $\{\pm 1\} \times \mu_2$ otherwise.*

*(f) On $A(F)$, $\beta(a(t_1), a(t_2)) = (t_1,t_2)_v$. Hence $\overline{A(F)}$ is abelian, and $(a(t),\zeta)^2 = (a(t^2),(t,-1)_v)$.*

*(g) If $g_1, g_2 \in SL(2,F)$ commute, then $\beta(g_1,g_2) = \beta(g_2,g_1)$. Consequently $\overline{H}$ is abelian for every abelian subgroup $H \subset SL(2,F)$; in particular $\overline{A(F)}$ and every $\overline{T_\theta}$ are abelian.*

*(h) Let $U = gNg^{-1}$ with $g \in SL(2,F)$. Then $u \mapsto (u,1)$ is the unique homomorphic section of $\overline{U} \to U$. Consequently, for every $(g,\zeta) \in \overline{SL(2,F)}$ and $u \in N(F)$,*

$$(g,\zeta)(u,1)(g,\zeta)^{-1} = (gug^{-1},1).$$

*Proof.* (a) For $n_1, n_2 \in N(F)$ all three values of $x$ equal 1.

(b) This is Kubota's computation [K2]; see also [Ge2]. The identity is also checked numerically for $F = \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7$, together with the existence of pairs in $K \times K$ with $\beta = -1$.

(c) Let $k_i = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \in K_m$ for $i = 1,2$, and $k_1k_2 = \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}$. All $a_i$ and $d_i$ lie in $1 + \varpi^m\mathcal{O} \subset F^{\times 2}$, and $c_3 = c_1a_2 + d_1c_2$. If $c_1, c_2, c_3 \neq 0$, then $c_1a_2/c_3 + d_1c_2/c_3 = 1$, so by (H2) and (H1)

$$\beta(k_1,k_2) = \left(\frac{c_3}{c_1}, \frac{c_3}{c_2}\right) = \left(\frac{c_1a_2}{c_3}, \frac{d_1c_2}{c_3}\right) = 1,$$

because $a_2$ and $d_1$ are squares. If $c_3 = 0 \neq c_1c_2$, then $x(k_1k_2) = d_3$ is a square and $c_2 = -c_1a_2/d_1$, so $\beta = (c_1,c_2) = (c_1,-c_1)(c_1,a_2/d_1) = 1$. If $c_1 = 0 \neq c_2$, then $x(k_1) = d_1$ is a square and $c_3 = d_1c_2$, so $\beta = (c_3/d_1,\, d_1) = 1$; the case $c_2 = 0 \neq c_1$ is the same with $a_2$ in place of $d_1$. If

$c_1 = c_2 = 0$, all $x$-values are squares. Finally, $1+\varpi\mathcal{O} \subset F^{\times 2}$ for non-dyadic $F$ by Hensel's lemma, and $1+4\varpi\mathcal{O} \subset F^{\times 2}$ for dyadic $F$ by the binomial series for $(1+4\varpi y)^{1/2}$; and $4\varpi\mathcal{O} = \varpi^{2e_v+1}\mathcal{O}$.

(d) At a complex place $\beta \equiv 1$ by (H3). For $t_1, t_2 > 0$, $(t_1, t_2)_{\mathbb{R}} = 1$ by (H3) and (f). The statement about $SO(2)$ is standard [Ge2].

(e) Commuting elements give commuting lifts by (g), so $\overline{\{\pm 1\}}$ is central; conversely, the image of a central element is central in $SL(2,F)$. Since $x(-1) = -1$ and $x(1) = 1$, $(-1,\zeta)^2 = (1, \beta(-1,-1)) = (1, (-1,-1)_v)$ by (2.1). A group of order 4 with an element of order 4 is cyclic, and one of exponent 2 is $\{\pm 1\} \times \mu_2$.

(f) Here $x(a(t)) = t^{-1}$, so (2.1) gives $\beta(a(t_1), a(t_2)) = (t_2^{-1}, t_1^{-1}) = (t_1, t_2)$ by (H1). The square is $(a(t^2), (t,t))$, and $(t,t) = (t,-1)$ by (H2).

(g) If $g_1 g_2 = g_2 g_1$, then $x(g_1 g_2) = x(g_2 g_1)$, and the first expression in (2.1) is symmetric in $g_1, g_2$ because the quadratic Hilbert symbol is symmetric (H1). By (2.2), lifts $(g_1, \zeta_1)$ and $(g_2, \zeta_2)$ commute if and only if $\beta(g_1, g_2) = \beta(g_2, g_1)$.

(h) If $g \in B(F)$, then $U = N$ and the claim is (a). Otherwise $g$ has lower left entry $\gamma \neq 0$, and $u_s = g\, n(s)\, g^{-1}$ has lower left entry $-\gamma^2 s$. So for $s_1, s_2 \neq 0$ with $s_1 + s_2 \neq 0$,

$$\beta(u_{s_1}, u_{s_2}) = \Big(\frac{s_1+s_2}{s_1}, \frac{s_1+s_2}{s_2}\Big) = \Big(\frac{s_1}{s_1+s_2}, \frac{s_2}{s_1+s_2}\Big) = 1$$

by (H1) and the Steinberg relation (H2). If $s_1 + s_2 = 0$, then $x(u_{s_1} u_{s_2}) = 1$ and $\beta = (1/s_1, -1/s_1) = 1$. So $u \mapsto (u, 1)$ is a homomorphism on $U$. Two homomorphic sections differ by a continuous homomorphism $U \cong F \to \mu_2$, which is trivial because $F = 2F$. Conjugation by $(g, \zeta)$ carries the section $n \mapsto (n, 1)$ of $\overline{N}$ to a homomorphic section of $\overline{gNg^{-1}}$, which is therefore $u \mapsto (u, 1)$. □

**Example 2.A.** Take $F = \mathbb{R}$. By (e) and (f), the element $(a(-1), 1) = (-1, 1)$ has order 4, so $\overline{\{\pm 1\}} \cong \mathbb{Z}/4$ and $\overline{A(\mathbb{R})}$ is an abelian group which is not $A(\mathbb{R}) \times \mu_2$. By (d), $\overline{A(\mathbb{R})}$ is generated by $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$ and $(-1, 1)$. At a non-dyadic place, on the other hand, $(-1,-1)_v = 1$, so the center is $\{\pm 1\} \times \mu_2$; and $\overline{A(\mathcal{O}^\times)}$ is split by $a \mapsto (a, 1)$, because $(t_1, t_2)_v = 1$ for units by (H3). This is consistent with (b), since $\kappa \equiv 1$ on $A(\mathcal{O}^\times)$.

*Remark* 2026.1. For $F = \mathbb{Q}_2$, Lemma 2.2(c) gives the splitting $k \mapsto (k, 1)$ over $K_3$, and $\beta$ is not identically 1 on $K_2 \times K_2$. The cover nevertheless splits over $K_2$, by $k \mapsto (k, (c,d)_2)$ for $c \neq 0$ and $k \mapsto (k, 1)$ for $c = 0$. This is the local form at 2 of the classical splitting over $\Gamma_1(4)$ [K2, Ge2]. We have verified it numerically only (`checks/sec2_n2_check.py`), and the same section fails on $K_1$. Nothing below depends on it: the rest of the paper uses only that some $K_m$ splits at each dyadic place.

**The cover of $G(\mathbb{Q}_p)$.** Let $\overline{G(\mathbb{Q}_p)}$ be the extension of $G(\mathbb{Q}_p) = \prod_{v|p} SL(2, k_v)$ by $\mu_2$ defined by the cocycle

$$\beta_p(g_1, g_2) = \prod_{v|p} \beta_v(g_{1,v}, g_{2,v}).$$

It is the *quotient* of $\prod_{v|p} \overline{SL(2,k_v)}$ by the kernel $\{(\zeta_v) \mid \prod_v \zeta_v = 1\}$ of the product map $\prod_{v|p} \mu_2 \to \mu_2$. (The 1995 notes call it a subgroup.) For $H \subset G(\mathbb{Q}_p)$ put $\overline{H} = p^{-1}(H)$. Let $B$, $N$, $A$ be the $\mathbb{Q}$-subgroups of $G$ obtained from those of $SL(2)$, and let $T = \prod_{v|p} T_v$ with each $T_v$ a maximal torus of $SL(2, k_v)$. Let $K_p = \prod_{v|p} K_v$. For $p$ odd, $K_p^* = \prod_{v|p} K_v^*$, with $K_v^*$ as in Lemma 2.2(b), maps to a subgroup of $\overline{G(\mathbb{Q}_p)}$ which we also denote by $K_p^*$.

**Lemma 2.3.** *(a) $n \mapsto (n,1)$ splits $\overline{N(\mathbb{Q}_p)}$.*

*(b) If $p$ is odd, $\overline{K_p} = K_p^* \times \mu_2$.*

*(c) For every $p$, $k \mapsto (k,1)$ splits $\overline{K_{m,p}}$, where $K_{m,p} = \prod_{v|p} K_{m,v}$, as soon as $m \geq 2e_v + 1$ for every $v \mid p$; if $p$ is odd, $m \geq 1$ suffices.*

*(d) If $p = \infty$, $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$, where $A(\mathbb{R})^0$ is the identity component of $A(k \otimes \mathbb{R})$.*

*(e) The center of $\overline{G(\mathbb{Q}_p)}$ is $\overline{\{\pm 1\}^{S_p}}$, where $S_p = \{v \mid p\}$. It is abelian, and the square of a lift of $\varepsilon = (\varepsilon_v)$ is $\prod_{v|p,\ \varepsilon_v = -1} (-1,-1)_v$.*

*(f) $\beta_p(a_1, a_2) = \prod_{v|p} (t_{1,v}, t_{2,v})_v$ for $a_i = a(t_i) \in A(\mathbb{Q}_p)$, and $\overline{A(\mathbb{Q}_p)}$ is abelian.*

*(g) The preimage of any abelian subgroup of $G(\mathbb{Q}_p)$, in particular $\overline{T}$, is abelian, and Lemma 2.2(h) holds for $G(\mathbb{Q}_p)$.*

*Proof.* Each statement holds componentwise by Lemma 2.2, and the quotient map is compatible with all of them. □

### 2.1.1 *NAK* decompositions

**Lemma 2.4.** *$\overline{G(\mathbb{Q}_p)} = N(\mathbb{Q}_p) \cdot \overline{A(\mathbb{Q}_p)} \cdot \overline{K_p}$, with $N(\mathbb{Q}_p)$ embedded by $n \mapsto (n,1)$. If $(n,1)\,\tilde{a}\,\tilde{k} = (n',1)\,\tilde{a}'\,\tilde{k}'$ are two such decompositions, then $p(\tilde{a}') \in p(\tilde{a})\,(A(\mathbb{Q}_p) \cap K_p)$. In particular $|t|_p := \prod_{v|p} |t_v|_v$ is well defined for $p(\tilde{a}) = a(t)$, and depends only on $(g,\zeta)$. For $p$ odd one may take $\tilde{k} \in K_p^*$.*

*Proof.* The Iwasawa decomposition $G(\mathbb{Q}_p) = N(\mathbb{Q}_p)A(\mathbb{Q}_p)K_p$ holds, with $a$ determined modulo $A(\mathbb{Q}_p) \cap K_p$, which is $\prod_{v|p} A(\mathcal{O}_v^\times)$ for $p < \infty$ and a product of copies of $\{\pm 1\}$ and of $\{\mathrm{diag}(z, \bar{z}) : |z| = 1\}$ for $p = \infty$. Since $\mu_2$ is central and lies in $\overline{A(\mathbb{Q}_p)}$, this lifts to $\overline{G(\mathbb{Q}_p)}$. For $p$ odd, $\overline{K_p} = K_p^* \mu_2$ and the $\mu_2$ factor can be moved into $\tilde{a}$. The absolute value $|t|_p$ is trivial on $A(\mathbb{Q}_p) \cap K_p$. □

### 2.1.2 Heights

For $g = (g_v) \in G(\mathbb{Q}_p)$ put $\|g\|_p = \prod_{v|p} \|g_v\|_v$, where $\|g_v\|_v = \max(|a|_v, |b|_v, |c|_v, |d|_v)$ for $v$ finite and $\|g_v\|_v = (|a|^2 + |b|^2 + |c|^2 + |d|^2)^{1/2}$, raised to the power $[k_v : \mathbb{R}]$, for $v$ archimedean (here $|\cdot|$ is the usual absolute value). Extend to the cover by $\|(g,\zeta)\|_p = \|g\|_p$.

Define $H = H_B \colon \overline{G(\mathbb{Q}_p)} \to \mathbb{R}$ by

$$H(g,\zeta) = \log |t|_p = \sum_{v|p} \log |t_v|_v, \tag{2.5}$$

where $(g,\zeta) = (n,1)\,\tilde{a}\,\tilde{k}$ as in Lemma 2.4 and $p(\tilde{a}) = a(t)$. This is the product of the normalized absolute values over $v \mid p$, that is $|N_{k \otimes \mathbb{Q}_p / \mathbb{Q}_p}(t)|_p$, and not the maximum used in the 1995 notes; see Section 2.1.3.

**Lemma 2.5.** *(a) $H(\tilde{a}\tilde{a}') = H(\tilde{a}) + H(\tilde{a}')$ for $\tilde{a}, \tilde{a}' \in \overline{A(\mathbb{Q}_p)}$.*

*(b) $H(\tilde{n}\, x\, \tilde{k}) = H(x)$ for $\tilde{n} \in \overline{N(\mathbb{Q}_p)}$, $x \in \overline{G(\mathbb{Q}_p)}$ and $\tilde{k} \in \overline{K_p}$.*

*(c) $|H(x)| \ll 1 + \log \|x\|_p$ for $x \in \overline{G(\mathbb{Q}_p)}$ [A2].*

The global analog, including invariance under $B(\mathbb{Q})$ on the left, is Lemma 2.8.

### 2.1.3 Characteristic functions of Weyl chambers

Let $P_0 = B$; the standard parabolic subgroups are $B$ and $G$. For a standard $P = MN_P$ let $A_P$ be the split component of the center of $M$ over $\mathbb{Q}$. Thus $A_G = 1$, because the center $\{\pm 1\}$ of $SL(2)$ is finite, and $A_B \cong \mathbb{G}_m$ is the image of $t \mapsto a(t)$, $t \in \mathbb{Q}^\times \subset k^\times$: the $\mathbb{Q}$-split part of the torus $A \cong \mathrm{Res}_{k/\mathbb{Q}}\,\mathbb{G}_m$, which has $\mathbb{Q}$-rank one.

The $\mathbb{Q}$-rational characters of $A$ form $X(A)_\mathbb{Q} \cong \mathbb{Z}$, generated by $\chi_1 = N_{k/\mathbb{Q}} \circ t$. Put $\mathfrak{a} = \mathfrak{a}_B = \mathrm{Hom}(X(A)_\mathbb{Q}, \mathbb{R})$, identified with $\mathbb{R}$ by $X \mapsto X(\chi_1)$. Then $H$ of (2.5) and its global analog are $\langle H(x), \chi_1\rangle = \log|\chi_1(\tilde{a})|$, which is why $H$ uses the norm. The simple root is $\alpha(a(t)) = t^2$; restricted to $A_B$ it is $\alpha = 2\chi_1$, so $\alpha(X) = 2X$ under $\mathfrak{a} \cong \mathbb{R}$. Let $\alpha^\vee = X_\alpha \in \mathfrak{a}$ be the coroot, so $X_\alpha = 1$ and $\alpha(\alpha^\vee) = 2$, and let $\varpi = \frac{1}{2}\alpha^\vee$ be the fundamental coweight, so $\alpha(\varpi) = 1$. (The 1995 notes call $\varpi$ the dual root.)

Let $\mathfrak{a}^+ = \{X \in \mathfrak{a} \mid \alpha(X) > 0\}$ and $(\mathfrak{a}^*)^+ = \{\Lambda \in \mathfrak{a}^* \mid \Lambda(\varpi) > 0\}$, with characteristic functions $\tau_B$ and $\hat{\tau}_B$, and set $\tau_G = \hat{\tau}_G = 1$. For standard $P_1 \subset P_2$ let $\tau_1^2$, $\hat{\tau}_1^2$ and $\sigma_1^2$ be as in [A2]. With $\mathfrak{a} \cong \mathbb{R}$:

$$\tau_B = \tau_B^G = \hat{\tau}_B = \hat{\tau}_B^G = \text{characteristic function of } \{u \in \mathbb{R} \mid u > 0\}, \qquad \tau_1^2 = \begin{cases} \tau_1, & P_2 = G, \\ 1, & P_2 \neq G, \end{cases} \tag{2.6}$$

$$\sigma_B^G = \tau_B, \qquad \sigma_B^B = 0, \qquad \sigma_G^G = 1. \tag{2.7}$$

### 2.1.4 Algebras of functions

We follow Li [Li I, Li IV]. A representation $\pi$ of a cover is *genuine* if $\pi(1, \zeta) = \zeta$ for $\zeta \in \mu_2$, and a function $f$ on the cover is *anti-genuine* if $f(g, \zeta) = \zeta^{-1} f(g, 1)$. Test functions are anti-genuine. Then $\pi(f) = \int f(\tilde{g})\pi(\tilde{g})\, d\tilde{g}$ picks up the factor $\int_{\mu_2} \zeta^{-1}\zeta\, d\zeta = 1$, so it is in general nonzero on genuine representations, and it vanishes on representations that are trivial on $\mu_2$.

For $n = 2$, $\zeta^{-1} = \zeta$, so "anti-genuine" and "genuine" define the same functions, and the conventions of the 1995 notes (genuine functions in §2.1.4, $\pi(g, \zeta) = \zeta^{-1}\pi(g, 1)$ in §3.1) agree with Li's. We keep Li's terminology so that the formulas remain correct for general $n$; see Remark 2026.4.

Let $C_c^\infty(\overline{G(\mathbb{Q}_p)})_-$ be the space of anti-genuine $f$ such that $g \mapsto f(g, 1)$ lies in $C_c^\infty(G(\mathbb{Q}_p))$. Let $p$ be odd. The *spherical Hecke algebra* $\mathcal{H}(\overline{G(\mathbb{Q}_p)}, K_p^*)$ consists of the $f \in C_c^\infty(\overline{G(\mathbb{Q}_p)})_-$ with $f(\tilde{k}\, x\, \tilde{k}') = f(x)$ for $\tilde{k}, \tilde{k}' \in K_p^*$. Its unit element is

$$e_p(\tilde{k}\zeta) = \zeta^{-1} \ (\tilde{k} \in K_p^*,\ \zeta \in \mu_2), \qquad e_p = 0 \text{ off } \overline{K_p}. \tag{2.8}$$

The function $(k, \zeta) \mapsto \zeta^{-1}$ on $\overline{K_p}$, which is what the identification $k \mapsto (k, 1)$ of the 1995 notes suggests, is $\prod_v \kappa_v \cdot e_p$; it is not bi-$K_p^*$-invariant.

### 2.1.5 Measures

Let $dk$ be the Haar measure on $K_p$ of total volume 1. Fix a nontrivial additive character $\psi_v$ of $k_v$ for each $v$, and let $dn$ be the measure on $N(\mathbb{Q}_p) \cong \bigoplus_{v|p} k_v$ that is the product of the self-dual measures for the $\psi_v$. Let $da$ be a Haar measure on $A(\mathbb{Q}_p)$. If $T$ is a maximal torus of $G$ over $\mathbb{Q}_p$, and $h \in G(\overline{\mathbb{Q}}_p)$ conjugates $T$ to $A$, let $dt$ be the pull-back of $da$ as in [R]; stably conjugate tori get compatible measures as in [R]. Put

$$\delta_B(a(t)) = |\alpha(a(t))|_p = |t|_p^2.$$

Normalize the Haar measure $dg$ on $G(\mathbb{Q}_p)$ by

$$\int_{G(\mathbb{Q}_p)} \phi(g)\,dg = \int_N \int_A \int_K \phi(nak)\,\delta_B(a)^{-1}\,dk\,da\,dn = \int_K \int_A \int_N \phi(kan)\,\delta_B(a)\,dn\,da\,dk. \tag{2.9}$$

The two expressions agree because $dg$ is invariant under $g \mapsto g^{-1}$ (as $G$ is unimodular) and $(nak)^{-1} = k^{-1}a^{-1}n^{-1}$; see [Kn, Ch. VIII] for the real case. (The 1995 notes have $\delta_B(a)^{1/2}$ for $g = kan$; the exponent must be 1 for $g = kan$ and $-1$ for $g = nak$.)

On a cover we use $d\tilde{g} = dg\,d\zeta$, where $d\zeta$ gives $\mu_2$ total volume 1. Then $\overline{K_p}$ has volume 1, and $e_p$ of (2.8) is idempotent: for $x = \tilde{k} \in K_p^*$,

$$(e_p * e_p)(x) = \int_{\overline{K_p}} e_p(y)\,e_p(y^{-1}x)\,dy = \int_{K_p^*} \int_{\mu_2} \zeta^{-1}\zeta\,d\zeta\,dy = 1.$$

## 2.2 Global aspects

The adelic cover is not the cover defined by $\prod_v \beta_v$: that product does not converge, because $\beta_v$ is not trivial on $K_v \times K_v$ (Lemma 2.2(b)). The standard remedy [K2, Ge2, KP], used here, is to correct $\beta_v$ by the coboundary of $\kappa_v$. This choice also fixes the splitting over $G(\mathbb{Q})$.

For each place $v$ of $k$ extend $\kappa_v$ to $SL(2,k_v)$ by

$$\kappa_v(g) = \begin{cases} (c,d)_v, & v \text{ finite non-dyadic},\ g \in K_v,\ 0 < |c|_v < 1, \\ 1, & \text{otherwise}, \end{cases}$$

so $\kappa_v \equiv 1$ at dyadic and archimedean places, and put

$$\sigma_v(g_1,g_2) = \beta_v(g_1,g_2)\,\kappa_v(g_1)\kappa_v(g_2)\kappa_v(g_1g_2). \tag{2.10}$$

This cocycle is cohomologous to $\beta_v$; it defines the same cover $\overline{SL(2,k_v)}$, identified with the $\beta_v$-model by $(g,\zeta)_\sigma \mapsto (g, \kappa_v(g)\zeta)_\beta$. By Lemma 2.2(b), $\sigma_v \equiv 1$ on $K_v \times K_v$ at every finite non-dyadic $v$. Since $g \in G(\mathbb{A})$ has $g_v \in K_v$ for almost all $v$, the product

$$\sigma_{\mathbb{A}}(g_1,g_2) = \prod_v \sigma_v(g_{1,v}, g_{2,v}) \tag{2.11}$$

has only finitely many factors different from 1. It is a continuous 2-cocycle on $G(\mathbb{A}) = SL(2,\mathbb{A}_k)$ and defines

$$1 \to \mu_2 \to \overline{G(\mathbb{A})} \xrightarrow{p} G(\mathbb{A}) \to 1.$$

Equivalently, $\overline{G(\mathbb{A})}$ is the restricted product of the $\overline{SL(2,k_v)}$ with respect to the subgroups $K_v^*$, divided by $\{(\zeta_v) \in \bigoplus_v \mu_2 \mid \prod_v \zeta_v = 1\}$. It is a *quotient* of the restricted product, not a subgroup. Factorization of genuine representations of $\overline{G(\mathbb{A})}$ (and of $\overline{A(\mathbb{A})}$ in Section 3) refers to this quotient. Group law, inverses and conjugation are given by (2.2) and (2.3) with $\sigma_{\mathbb{A}}$ in place of $\beta$. For $H \subset G(\mathbb{A})$ put $\overline{H} = p^{-1}(H)$.

For $\gamma \in SL(2,k)$ we have $\gamma \in K_v$ and $\mathrm{ord}_v(c) \le 0$ or $c = 0$ for almost all $v$, so

$$\kappa_{\mathbb{A}}(\gamma) = \prod_v \kappa_v(\gamma) = \prod_{\substack{v \text{ finite non-dyadic} \\ \gamma \in K_v,\ 0<|c|_v<1}} (c,d)_v \tag{2.12}$$

is a finite product.

**Lemma 2.6.** *(a) The map $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$ is a homomorphism $G(\mathbb{Q}) \to \overline{G(\mathbb{A})}$, so $\overline{G(\mathbb{Q})} = G(\mathbb{Q}) \times \mu_2$.*

*(b) $n \mapsto (n, 1)$ splits $\overline{N(\mathbb{A})}$, and on $B(\mathbb{Q})$ the splitting of (a) is $b \mapsto (b, 1)$.*

*(c) Let $S$ be a finite set of places containing the archimedean and dyadic places, and let $K_f = \prod_{v \in S \smallsetminus \Sigma_k^\infty} K_{m_v,v} \times \prod_{v \notin S} K_v$, with $m_v \geq 1$, and $m_v \geq 2e_v + 1$ if $v$ is dyadic. Then $k \mapsto (k, 1)$ is a homomorphism $K_f \to \overline{G(\mathbb{A})}$; we write $K_f^*$ for its image. (At $v \notin S$, translated to the $\beta_v$-model, its component is $K_v^*$.)*

*(d) The center of $\overline{G(\mathbb{A})}$ is $\overline{Z(G(\mathbb{A}))}$, where $Z(G(\mathbb{A})) = \prod_v \{\pm 1\}$. It is abelian, and a lift of $\varepsilon = (\varepsilon_v)$ has square $\prod_{v:\, \varepsilon_v = -1} (-1, -1)_v$, a finite product. It contains the subgroup $Z(G(\mathbb{Q})) \times \mu_2$ of order 4, a direct product, embedded by (a). It is not the direct product $Z(G(\mathbb{A})) \times \mu_2$ when $k$ has a real place.*

*(e) $\sigma_{\mathbb{A}}(a(t_1), a(t_2)) = \prod_v (t_{1,v}, t_{2,v})_v$ for $t_1, t_2 \in \mathbb{A}_k^\times$, and the preimage of every abelian subgroup of $G(\mathbb{A})$, in particular $\overline{A(\mathbb{A})}$, is abelian.*

*Proof.* (a) For $\gamma_1, \gamma_2 \in SL(2, k)$, $\prod_v \beta_v(\gamma_1, \gamma_2) = 1$ by the product formula (H4), because $\beta_v(\gamma_1, \gamma_2)$ is the local Hilbert symbol of two elements of $k^\times$. Hence $\sigma_{\mathbb{A}}(\gamma_1, \gamma_2) = \kappa_{\mathbb{A}}(\gamma_1)\kappa_{\mathbb{A}}(\gamma_2)\kappa_{\mathbb{A}}(\gamma_1\gamma_2)$, which is the condition for $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$ to be a homomorphism.

(b) On $N$ every $\beta_v$ and $\kappa_v$ is 1. On $B(\mathbb{Q})$ every $\kappa_v$ is 1 because $c = 0$.

(c) It suffices that $\sigma_v \equiv 1$ on the $v$-component of $K_f \times K_f$ for each finite $v$. At $v \notin S$ this is Lemma 2.2(b). At finite $v \in S$, $\kappa_v \equiv 1$ on $K_{m_v,v}$ (non-dyadic: $c \in \varpi\mathcal{O}$ and $d$ is a square; dyadic: by definition), so $\sigma_v = \beta_v \equiv 1$ there by Lemma 2.2(c).

(d) Lemma 2.2(g) applies to $\sigma_v$, since $\kappa_v(g_1)\kappa_v(g_2)\kappa_v(g_1g_2)$ is symmetric in commuting $g_1, g_2$; so preimages of abelian subgroups are abelian, and $\overline{Z(G(\mathbb{A}))}$ is central. Its image under $p$ is central, so equality holds. Since $\kappa_v(\pm 1) = 1$, the square of a lift of $\varepsilon$ is $\prod_v \sigma_v(\varepsilon_v, \varepsilon_v) = \prod_{\varepsilon_v = -1} (-1, -1)_v$, which is finite by (H3). For $\varepsilon = -1 \in Z(G(\mathbb{Q}))$ it is $\prod_v (-1, -1)_v = 1$ by (H4). If $v_0$ is real, the lift of the element that is $-1$ at $v_0$ and 1 elsewhere has order 4.

(e) As in Lemma 2.2(f), since $\kappa_v \equiv 1$ on $A$. □

From now on we identify $G(\mathbb{Q})$ with its image under Lemma 2.6(a), and $N(\mathbb{A})$ with $N(\mathbb{A}) \times 1$. By (b), the two identifications agree on $N(\mathbb{Q})$. The splitting over $B(\mathbb{Q})$ by $b \mapsto (b, 1)$ is what the constant terms, Eisenstein series and the truncation operator of Sections 4 and 5 use.

**Example 2.B.** Let $k = \mathbb{Q}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$ with $c \neq 0$. Then $\gamma \in K_v$ for every $v$, and for odd $p \mid c$, $(c, d)_p = \left(\frac{d}{p}\right)^{\mathrm{ord}_p(c)}$ by (H3). So

$$\kappa_{\mathbb{A}}(\gamma) = \left(\frac{d}{c'}\right), \qquad c' = \text{the odd part of } |c|,$$

a Jacobi symbol: the factor behind the theta multiplier on $\Gamma_0(4)$ [K2, Ge2]. The script checks this on random elements of $SL(2, \mathbb{Z})$.

*Remark* 2026.2. The 1995 notes define the splitting as $\gamma \mapsto (\gamma, s_{\mathbb{A}}(\gamma)^{-1})$, with $s_v(\gamma)^{-1} = (c, d)_v$ when $v$ is finite, $cd \neq 0$ and $n \nmid \mathrm{ord}_v(c)$, and with the cocycle $\prod_v \beta_v$. That cocycle does not converge, and the formula belongs to a different normalization. In the normalization (2.11), $\kappa_{\mathbb{A}}$ of (2.12) replaces $s_{\mathbb{A}}^{-1}$. For $n = 2$ the two agree at every non-dyadic $v$ with $\gamma \in K_v$: if $\mathrm{ord}_v(c) > 0$

is even, or if $\mathrm{ord}_v(c) = 0$, both $(c,d)_v$ and $\kappa_v(\gamma)$ are 1. They differ only at the finitely many places where $\gamma \notin K_v$, and at dyadic places.

Let $K_0 = \prod_v K_v$ be the standard maximal compact subgroup of $G(\mathbb{A})$, with $K_v$ as in Section 2.1; $K_f$ and $K_\infty$ denote its finite and archimedean parts.

**Lemma 2.7** (Strong approximation; compare [K2, Prop. 5], [Ge2])**.** *Let $K_f$ and $K_f^*$ be as in Lemma 2.6(c), and put $\Gamma = G(\mathbb{Q}) \cap K_f$, a congruence subgroup of $SL(2,\mathcal{O}_k)$, viewed in $G(\mathbb{R}) = SL(2, k\otimes\mathbb{R})$. For $\gamma \in \Gamma$ let $\tilde{k}_\gamma \in K_f^*$ be the lift of $\gamma_f$. Then $\tilde{\gamma} = \gamma\,\tilde{k}_\gamma^{-1}$ lies in $\overline{G(\mathbb{R})}$, the map $\gamma \mapsto \tilde{\gamma}$ is an injective homomorphism, its image $\widetilde{\Gamma}$ is discrete of finite covolume, and*

$$G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}/K_f^* \;\cong\; \widetilde{\Gamma}\backslash\overline{G(\mathbb{R})}.$$

*Proof.* Since $SL(2)$ is simply connected and $G(\mathbb{R})$ is not compact, strong approximation gives $G(\mathbb{A}) = G(\mathbb{Q})\,G(\mathbb{R})\,K_f$. Lifting to the cover, with $\mu_2 \subset \overline{G(\mathbb{R})}$, gives $\overline{G(\mathbb{A})} = G(\mathbb{Q})\,\overline{G(\mathbb{R})}\,K_f^*$, so the double coset space has a single component. The stabilizer computation is the usual one: $\gamma \in G(\mathbb{Q})$ lies in $\overline{G(\mathbb{R})}K_f^*$ exactly when $\gamma_f \in K_f$, and then $\gamma = \tilde{\gamma}\,\tilde{k}_\gamma$. The map $\gamma \mapsto \tilde{\gamma}$ is a homomorphism because lifts of elements of $G(\mathbb{A})$ with disjoint supports commute: $\sigma_v(1,h) = \sigma_v(h,1) = 1$. □

### 2.2.1 *NAK* decompositions

**Lemma 2.8.** *(a) $\overline{G(\mathbb{A})} = N(\mathbb{A})\cdot\overline{A(\mathbb{A})}\cdot\overline{K_0}$. In a decomposition $(n,1)\,\tilde{a}\,\tilde{k}$, $p(\tilde{a})$ is determined modulo $A(\mathbb{A})\cap K_0$, and*

$$H(g,\zeta) = \log|t|_{\mathbb{A}_k} = \sum_v \log|t_v|_v, \qquad p(\tilde{a}) = a(t),$$

*is well defined.*

*(b) $H$ satisfies Lemma 2.5(a)–(c) globally, and $H(\gamma x) = H(x)$ for $\gamma \in B(\mathbb{Q})$ and $x \in \overline{G(\mathbb{A})}$.*

*Proof.* (a) As for Lemma 2.4, place by place. (b) For $\gamma = n'a(s) \in B(\mathbb{Q})$, identified with $(\gamma,1)$ by Lemma 2.6(b), and $x = (n,1)\tilde{a}\tilde{k}$, $\gamma x = (n'',1)\,(a(s),1)\tilde{a}\,\tilde{k}$ with $n'' \in N(\mathbb{A})$, so $H(\gamma x) = \log|s|_{\mathbb{A}_k} + H(x) = H(x)$ by the product formula. □

### 2.2.2 Heights

For $g = (g_v) \in G(\mathbb{A})$ put $\|g\| = \prod_v \|g_v\|_v$ and $\|(g,\zeta)\| = \|g\|$.

**Lemma 2.9.** *For $x, y \in \overline{G(\mathbb{A})}$ and $\tilde{k}, \tilde{k}' \in \overline{K_0}$: (a) $\|x\| \geq 1$; (b) $\|\tilde{k}\,x\,\tilde{k}'\| = \|x\|$; (c) $\|xy\| \leq \|x\|\,\|y\|$; (d) $\|x^{-1}\| = \|x\|$.*

*Proof.* Each holds for $G(\mathbb{A})$ place by place, since $\det g_v = 1$ forces $\|g_v\|_v \geq 1$. For (d), $g^{-1}$ has the same entries as $g$ up to order and sign. The cover does not change $\|\cdot\|$. □

### 2.2.3 Algebras of functions

A function on $\overline{G(\mathbb{A})}$ is *anti-genuine* if $f(g,\zeta) = \zeta^{-1}f(g,1)$. Let $\mathcal{H}(\overline{G(\mathbb{A})})$ be the span of the functions $f = \bigotimes_v f_v$, $f_v \in C_c^\infty(\overline{SL(2,k_v)})_-$, with $f_v$ equal to the unit element $e_v$ of (2.8) (defined with $K_v^*$) for almost all $v$. Such a product is a function on the restricted product of the local covers which is invariant under $\{(\zeta_v) \mid \prod\zeta_v = 1\}$, because each $f_v$ is anti-genuine; so it descends to $\overline{G(\mathbb{A})}$ and is anti-genuine there.

### 2.2.4 Measures

Let $dk = \prod_v dk_v$ on $K_0$. Let $dg = \prod_v dg_v$ be the Tamagawa measure on $G(\mathbb{A})$ [W2], with $dg_v$ built from the self-dual measures of a fixed nontrivial character $\psi = \prod \psi_v$ of $\mathbb{A}_k/k$; then $\mathrm{vol}(G(\mathbb{Q})\backslash G(\mathbb{A})) = 1$. For a maximal torus $T$ of $G$ over $\mathbb{Q}$ let $dt$ be the Tamagawa measure on $T(\mathbb{A})$, written $da$ for $T = A$, and let $dn$ be the product measure on $N(\mathbb{A})$, for which $\mathrm{vol}(N(\mathbb{Q})\backslash N(\mathbb{A})) = 1$. On the cover put $d\tilde{g} = dg\, d\zeta$ with $\mathrm{vol}(\mu_2) = 1$, so $\mathrm{vol}(G(\mathbb{Q})\backslash \overline{G(\mathbb{A})}) = 1$.

## 2.3 The double cover of $GL(2)$

Section 7 compares $SL(2)$-conjugacy with $GL(2)$-conjugacy on the cover, so we need a double cover of $GL(2)$ which restricts to the one above. There are two natural choices for $n = 2$ (Remark 2026.3); we fix one, and compute the conjugation action of $GL(2,F)$ on $\overline{SL(2,F)}$ for later use.

For $g_1, g_2 \in GL(2,F)$, with $x(g)$ defined as before, put

$$\sigma(g_1,g_2) = \Big(\frac{x(g_1g_2)}{x(g_1)},\ \frac{x(g_1g_2)}{x(g_2)\det g_1}\Big)_v, \tag{2.13}$$

the $n = 2$ case of Kubota's cocycle on $GL(2)$ [K2, Ge2]. Let $\overline{GL(2,F)}$ be the corresponding extension by $\mu_2$.

**Lemma 2.10.** *(a) $\sigma$ is a 2-cocycle on $GL(2,F)$ whose restriction to $SL(2,F)$ is $\beta$. Hence $p^{-1}(SL(2,F)) \subset \overline{GL(2,F)}$ is the cover $\overline{SL(2,F)}$ of Section 2.1.*

*(b) For $v$ finite non-dyadic, $k \mapsto (k, \kappa'(k))$ is a homomorphism on $GL(2,\mathcal{O})$, where $\kappa'(k) = (c,\, d/\det k)_v$ if $0 < |c|_v < 1$ and $\kappa'(k) = 1$ otherwise. On $SL(2,\mathcal{O})$, $\kappa' = \kappa$.*

*(c) Conjugation by any lift of $h \in GL(2,F)$ defines an automorphism $\mathrm{Ad}(h)$ of $\overline{SL(2,F)}$ lifting conjugation by $h$, and $h \mapsto \mathrm{Ad}(h)$ is an action of $GL(2,F)$. If $h$ commutes with $\gamma \in SL(2,F)$, then*

$$\mathrm{Ad}(h)(\gamma,\zeta) = \Big(\gamma,\ \Big(\frac{x(h\gamma)}{x(h)},\ \det h\Big)_v \zeta\Big). \tag{2.14}$$

*In particular $\mathrm{Ad}(\mathrm{diag}(y,1))(a(t),\zeta) = (a(t),(y,t)_v\zeta)$, so the preimage of the diagonal torus of $GL(2,F)$ is not abelian unless $F = \mathbb{C}$. Likewise $E^\times \supset T_\theta$ (embedded as $\{\begin{pmatrix} a & b\theta \\ b & a \end{pmatrix}\}$) acts on $\overline{T_\theta}$ through (2.14).*

*Proof.* (a) On $SL(2)$, $\det g_1 = 1$ and (2.13) becomes (2.1). The cocycle identity is due to Kubota [K2]; we have also checked it numerically.

(b) We have checked this numerically for $F = \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7$; see Remark 2026.5.

(c) Since $\mu_2$ is central, conjugation by $(h,\zeta)$ does not depend on $\zeta$, and it preserves $p^{-1}(SL(2,F))$. If $h\gamma = \gamma h$, then $\mathrm{Ad}(h)(\gamma,\zeta) = (\gamma, \zeta\,\sigma(h,\gamma)\sigma(\gamma,h))$. With $x_3 = x(h\gamma)$, bimultiplicativity gives $\sigma(h,\gamma) = (x_3/x(h), x_3/x(\gamma))\,(x_3/x(h), \det h)$, and $\sigma(\gamma,h) = (x_3/x(\gamma), x_3/x(h))$ since $\det\gamma = 1$. By symmetry (H1) the first factors cancel. For $h = \mathrm{diag}(y,1)$ and $\gamma = a(t)$: $x(h\gamma) = t^{-1}$ and $x(h) = 1$, giving $(t^{-1},y) = (y,t)$. This is nontrivial for suitable $y,t$ unless $F = \mathbb{C}$. □

The consequences for Section 7 (the functions $\delta \mapsto I_{\mathcal{O}}(f^\delta)$ are defined on $D(T/F)$ only up to the character (2.14), and the choice of $D(A/F)$) are taken up there.

*Remark* 2026.3. The cocycle $\sigma'(g_1,g_2) = \sigma(g_1,g_2)\,(\det g_1, \det g_2)_v$ is another double cover of $GL(2,F)$ with the same restriction to $SL(2,F)$; these are the two Kazhdan–Patterson covers for $n=2$ [KP]. We fix $\sigma$. Whether Section 7 needs $\sigma'$ instead has not been checked at this stage. The commutator formula (2.14) is the same for $\sigma'$: the extra factor $(\det h, \det\gamma)(\det\gamma, \det h)$ is 1 by (H1). The adelic cover of $GL(2,\mathbb{A}_k)$ is built as in Section 2.2, with $\kappa'_v$ in place of $\kappa_v$; Hilbert reciprocity for $\prod_v \sigma_v$ on $GL(2,k)$ is checked numerically for $k=\mathbb{Q}$.

*Remark* 2026.4. For general $n$ (with $\mu_n \subset k$) the following change. The conjugation action of $\overline{A(F)}$ on itself is through $(t_1,t_2)_n^2$, so $\overline{A(F)}$ is abelian only for $n \le 2$; the argument of Lemma 2.2(g) uses the symmetry of the quadratic symbol and does not extend. Lemma 2.2(c) holds with $1+\varpi^m\mathcal{O} \subset F^{\times n}$ (the same proof). Genuine ($\zeta$) and anti-genuine ($\zeta^{-1}$) differ, and with the 1995 convention $R(f)$ vanishes on $L^2_{\mathrm{gen}}$ unless $n \mid 2$.

*Remark* 2026.5. The script `checks/sec2_n2_check.py` verifies, in exact rational arithmetic at $v = \infty, 2, 3, 5, 7$: the cocycle identity and the two forms of (2.1); Lemma 2.2(a)–(c), (e)–(h); Remark 2026.1; the product formula and the splitting of Lemma 2.6(a), (b), (d); Example 2.B; and Lemma 2.10(a)–(c) with Remark 2026.3. The checks are on random elements of $SL(2,\mathbb{Q})$, $GL(2,\mathbb{Q})$, $SL(2,\mathbb{Z}_{(p)})$ and the tori $T_\theta(\mathbb{Q})$; they are evidence, not proofs. Of the statements in this section, Lemma 2.2(b) and Lemma 2.10(a), (b) are supported here only by citation and these checks.

# 3 The rank 0 case

Before the trace formula for $\overline{G(\mathbb{A})}$ we treat the covers of the tori of $G$, where it is elementary. For $n=2$ these covers are abelian (Lemma 2.6(e)), but in general not split, so a genuine automorphic character cannot be written as an ordinary one times a sign (Example 3.A). The trace formula is then Poisson summation twisted by one fixed genuine automorphic character (Theorem 3.4, Lemma 3.5), and it gives a bijection between genuine automorphic characters and Hecke characters (Theorem 3.2). For the split torus, Weil's character $\chi_\psi$ (Lemma 3.3) makes the bijection canonical once $\psi$ is fixed. We also define the distributions $I^A_{\mathcal{O}}$ and $I^A_{\tilde{\chi}}$ used in Section 6. We work with an arbitrary maximal torus because Section 7 applies the same formula to anisotropic tori.

## 3.1 The cover of a torus

Let $T$ be a maximal torus of $G$ defined over $\mathbb{Q}$. Then $T = \mathrm{Res}_{k/\mathbb{Q}} T_k$ for a maximal torus $T_k$ of $SL(2)$ over $k$, which is either split, and then $GL(2,k)$-conjugate to $A$, or of the form $T_\theta$ of Section 2.1 with $E = k(\sqrt{\theta})$ a quadratic extension of $k$. Put

$$T(\mathbb{A})^1 = \{t \in T(\mathbb{A}) \mid |\chi(t)| = 1 \text{ for all } \chi \in X(T)_{\mathbb{Q}}\}.$$

For $T=A$ this is $A(\mathbb{A})^1 = \{a(t) \mid |t|_{\mathbb{A}_k} = 1\}$, the kernel of $H$ of Lemma 2.8; for $T_k = T_\theta$ the torus $T$ is anisotropic over $\mathbb{Q}$ and $T(\mathbb{A})^1 = T(\mathbb{A})$. In both cases $T(\mathbb{Q}) \subset T(\mathbb{A})^1$ and $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ is compact [PR].

Let $\overline{T(\mathbb{A})}$ and $\overline{T(\mathbb{A})^1}$ be the preimages in $\overline{G(\mathbb{A})}$, and $\overline{T(k_v)}$ the preimage of $T_k(k_v)$ in $\overline{SL(2,k_v)}$. By Lemmas 2.2(g) and 2.6(e) these groups are abelian. On $A$ the cocycle is

$$\sigma_{\mathbb{A}}(a(t_1),a(t_2)) = \prod_v (t_{1,v}, t_{2,v})_v, \qquad t_1, t_2 \in \mathbb{A}_k^\times, \tag{3.1}$$

since $\kappa_v \equiv 1$ on $A$. (The 1995 formula $\beta(g_1,g_2) = (g_1^{-1},g_2^{-1})(-g_1g_2^{-1},g_1^{-1}g_2^{-1})$ reduces to this for $n=2$.) Exactly as for $G$ in Section 2.2, $\overline{T(\mathbb{A})}$ is the *quotient* of the restricted product of the $\overline{T(k_v)}$, taken with respect to the subgroups $T(\mathcal{O}_v)^* = \{(t,\kappa_v(t)) \mid t \in T_k(k_v) \cap K_v\}$, by $\{(\zeta_v) \mid \prod_v \zeta_v = 1\}$; it is not a subgroup of that restricted product. Each $\overline{T(k_v)}$ maps injectively into $\overline{T(\mathbb{A})}$. We regard $T(\mathbb{Q})$ as a subgroup of $\overline{T(\mathbb{A})^1}$ by Lemma 2.6(a), $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$; for $T=A$ this is $a(t) \mapsto (a(t),1)$, $t \in k^\times$.

**Conventions.** As in Section 2.1.4, a representation $\pi$ of one of these covers is *genuine* if $\pi(1,\zeta) = \zeta$, and a function $f$ is *anti-genuine* if $f(x\zeta) = \zeta^{-1}f(x)$ for $\zeta \in \mu_2$. For $n=2$ the two conditions on functions coincide. Let $C_c^\infty(\overline{T(k_v)})_-$ and $C_c^\infty(\overline{T(\mathbb{A})^1})_-$ be the anti-genuine functions whose restriction to $T \times \{1\}$ is smooth and compactly supported. At places where $T(\mathcal{O}_v)^*$ is defined let $e_{T,v}$ be the anti-genuine function equal to 1 on $T(\mathcal{O}_v)^*$ and supported on $\overline{T(\mathcal{O}_v)}$. Let $\mathcal{H}(\overline{T(\mathbb{A})})$ be the span of the products $\bigotimes_v f_v$, $f_v \in C_c^\infty(\overline{T(k_v)})_-$, with $f_v = e_{T,v}$ for almost all $v$; these descend to $\overline{T(\mathbb{A})}$ as in Section 2.2. Measures on $T(\mathbb{A})$ are as in Section 2.2, and on covers $\mu_2$ has total volume 1.

For a locally compact group $H$ let $\Pi(H)$ be the set of (equivalence classes of) irreducible admissible genuine representations of $H$, and $\Pi_u(H)$ the unitary ones.

**Lemma 3.1.** *Let $F = k_v$.*

*(a) Every element of $\Pi(\overline{T(F)})$, $\Pi(\overline{T(\mathbb{A})})$ or $\Pi(\overline{T(\mathbb{A})^1})$ is a genuine character.*

*(b) Genuine characters of $\overline{T(F)}$ exist, and any two differ by a character of $T(F)$: they form a torsor under $T(F)^\wedge$.*

*(c) Every genuine character $\bar\chi$ of $\overline{T(\mathbb{A})}$ factors uniquely as $\bar\chi = \prod_v \bar\chi_v$, where $\bar\chi_v$ is a genuine character of $\overline{T(k_v)}$, trivial on $T(\mathcal{O}_v)^*$ for almost all $v$. Conversely, every such family defines a genuine character of $\overline{T(\mathbb{A})}$. The same holds for unitary characters.*

*Proof.* (a) The groups are abelian, so Schur's lemma applies; for unitary representations of locally compact abelian groups this is standard, and for admissible representations Schur's lemma holds because they have countable dimension. Since $\mu_2$ is central, it acts by a character, and that character is the identity because the representation is genuine.

(b) $\mu_2$ is a closed subgroup of the locally compact abelian group $\overline{T(F)}$, so the character $\zeta \mapsto \zeta$ of $\mu_2$ extends [Ro]. The quotient of two genuine characters is trivial on $\mu_2$, hence is the pull-back of a character of $T(F)$.

(c) Put $\bar\chi_v = \bar\chi \circ \iota_v$, where $\iota_v$ is the inclusion of $\overline{T(k_v)}$. Since $\bar\chi$ is continuous, its kernel contains an open subgroup, and hence contains $T(\mathcal{O}_v)^*$ for all $v$ outside a finite set $S$. Every $x \in \overline{T(\mathbb{A})}$ is the image of some $(x_v)$ with $x_v \in T(\mathcal{O}_v)^*$ for $v \notin S'$, and then $\bar\chi(x) = \prod_{v\in S'} \bar\chi_v(x_v)$. Conversely, $\prod_v \bar\chi_v$ is a character of the restricted product, and it is trivial on $\{(\zeta_v) \mid \prod \zeta_v = 1\}$ because each $\bar\chi_v$ is genuine. So it descends to $\overline{T(\mathbb{A})}$. □

This replaces Lemma 3.1 of the 1995 notes, which cited [Ge2] for factorizability of genuine representations of the restricted product. For $n=2$ the statement concerns the quotient and needs no citation.

Let $X_T = T(\mathbb{Q})\backslash\overline{T(\mathbb{A})^1}$. It is a compact abelian group, an extension of the compact group $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ by $\mu_2$; $\mu_2$ injects into $X_T$ because $T(\mathbb{Q})$ meets $\mu_2$ trivially in $\overline{G(\mathbb{A})}$. Let $\Pi_{\mathrm{aut}}(X_T)$ be the set of genuine characters of $\overline{T(\mathbb{A})^1}$ that are trivial on $T(\mathbb{Q})$, i.e. the genuine characters of $X_T$.

**Theorem 3.2.** *(a)* $L^2(X_T)_{\mathrm{gen}} = \widehat{\bigoplus}_{\bar\chi\in\Pi_{\mathrm{aut}}(X_T)} \mathbb{C}\,\bar\chi$, *each character occurring with multiplicity one.*

*(b)* $\Pi_{\mathrm{aut}}(X_T)$ *is not empty. For any* $\bar\chi_0 \in \Pi_{\mathrm{aut}}(X_T)$, *the map* $\chi \mapsto \bar\chi_0\cdot(\chi\circ p)$ *is a bijection from the characters of* $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ *onto* $\Pi_{\mathrm{aut}}(X_T)$. *Replacing* $\bar\chi_0$ *by another element changes this bijection by a translation of* $(T(\mathbb{Q})\backslash T(\mathbb{A})^1)^\wedge$.

*(c) For* $T = A$, *the character* $\chi_\psi$ *of Lemma 3.3 lies in* $\Pi_{\mathrm{aut}}(X_A)$; *it depends only on the fixed additive character* $\psi$ *of* $\mathbb{A}_k/k$.

*Proof.* (a) Peter–Weyl for the compact abelian group $X_T$: $L^2(X_T)$ is the Hilbert sum of its characters, each once, and the genuine subspace is spanned by the genuine ones. (b) Apply Lemma 3.1(b) to the closed subgroup $\mu_2$ of $X_T$; the second sentence follows as in that lemma. (c) is Lemma 3.3(c). □

Theorem 3.2 is the $n = 2$ form of Theorem 3.2 of the 1995 notes, restated as a correspondence of automorphic spectra. The 1995 statement matches genuine representations with characters $\chi^N$, $N = n/\gcd(n,2)$; for $n = 2$, $N = 1$, and the matching requires the choice of $\bar\chi_0$, which the 1995 text did not make. The choice cannot be avoided:

**Example 3.A.** Let $k = \mathbb{Q}$, and let $\varepsilon \in A(\mathbb{A})^1$ be $a(-1)$ at the real place and 1 at every finite place. By Lemma 2.2(e) any lift $\tilde\varepsilon$ satisfies $\tilde\varepsilon^2 = (1,(-1,-1)_\infty) = (1,-1)$. So every genuine character $\bar\chi$ of $\overline{A(\mathbb{A})^1}$ has $\bar\chi(\tilde\varepsilon) = \pm i$, and no genuine character is of the form $(\chi\circ p)\cdot\mathrm{sgn}$ for a splitting $A(\mathbb{A})^1 \to \overline{A(\mathbb{A})^1}$; indeed no such splitting exists.

For the split torus there is a canonical choice, given by Weil indices [We64]. Let $\psi = \prod_v \psi_v$ be the nontrivial character of $\mathbb{A}_k/k$ fixed in Section 2.2. For $a \in k_v^\times$ let $\gamma_v(a) \in \mu_8$ be the Weil index of the character $x \mapsto \psi_v(ax^2)$ of $k_v$, normalized by

$$\int_{k_v} \psi_v(ax^2)\,dx = \gamma_v(a)\,|2a|_v^{-1/2} \tag{3.2}$$

(self-dual measure; for $v$ finite the integral is the limit over the lattices $\varpi^{-M}\mathcal{O}_v$, $M \to \infty$).

**Lemma 3.3** (Weil [We64])**.** *(a)* $\gamma_v(a)$ *depends only on the class of* $a$ *in* $k_v^\times/k_v^{\times 2}$, *and* $\gamma_v(ab)\gamma_v(1) = \gamma_v(a)\gamma_v(b)(a,b)_v$.

*(b) If* $v$ *is finite and non-dyadic and* $\psi_v$ *is unramified, then* $\gamma_v(u) = \gamma_v(1) = 1$ *for* $u \in \mathcal{O}_v^\times$. *For* $a \in k^\times$, $\prod_v \gamma_v(a) = 1$.

*(c) Hence* $\chi_{\psi,v}(a(t),\zeta) = \zeta\,\gamma_v(t)/\gamma_v(1)$ *is a genuine character of* $\overline{A(k_v)}$, *trivial on* $A(\mathcal{O}_v)^* = A(\mathcal{O}_v)\times\{1\}$ *for almost all* $v$, *and* $\chi_\psi = \prod_v \chi_{\psi,v}$ *is a genuine character of* $\overline{A(\mathbb{A})}$ *trivial on* $A(\mathbb{Q})$. *Its restriction to* $\overline{A(\mathbb{A})^1}$ *lies in* $\Pi_{\mathrm{aut}}(X_A)$.

*(d) For* $b \in k^\times$ *and* $\psi_b(x) = \psi(bx)$, $\chi_{\psi_b} = \chi_\psi\cdot\big(a(t)\mapsto\prod_v(b,t_v)_v\big)$.

*Proof.* (a) and (b) are due to Weil [We64, §§14, 28]. (c) By (a), $\chi_{\psi,v}(a(t_1),1)\chi_{\psi,v}(a(t_2),1) = (t_1,t_2)_v\,\chi_{\psi,v}(a(t_1t_2),1) = \chi_{\psi,v}((a(t_1),1)(a(t_2),1))$, using the group law (3.1). Unramifiedness is (b), and triviality on $(a(t),1)$, $t \in k^\times$, is the product formula in (b) applied to $\gamma(t)$ and to $\gamma(1)$. (d) $\psi_{b,v}(ax^2) = \psi_v(abx^2)$, so $\gamma_{\psi_b,v}(t)/\gamma_{\psi_b,v}(1) = \gamma_v(bt)/\gamma_v(b) = (\gamma_v(t)/\gamma_v(1))\,(b,t)_v$ by (a). □

**Example 3.B.** Let $k=\mathbb{Q}$, $\psi_p(x)=e^{2\pi i\{x\}_p}$ and $\psi_\infty(x)=e^{-2\pi i x}$, so $\psi$ is trivial on $\mathbb{Q}$. Then $\gamma_\infty(a)=e^{-\pi i\,\mathrm{sgn}(a)/4}$, and for odd $p$, $\gamma_p(u)=1$ and $\gamma_p(pu)=\left(\frac{u}{p}\right)\epsilon_p$ for $u\in\mathbb{Z}_p^\times$, where $\epsilon_p=1$ if $p\equiv 1 \pmod 4$ and $\epsilon_p=i$ if $p\equiv 3 \pmod 4$. So $\chi_{\psi,\infty}(a(-1),1)=i$, in accordance with Example 3.A, and $\chi_{\psi,p}(a(p),1)=\epsilon_p$ for odd $p$. The dyadic values are tabulated by the check script, which computes $\gamma_v$ from (3.2) as a finite Gauss sum and verifies Lemma 3.3(a)–(c) at $v=\infty,2,3,5,7$ and the product formula on random $a\in\mathbb{Q}^\times$.

### 3.2 The trace formula

Let $R$ be the right regular representation of $\overline{T(\mathbb{A})^1}$ on $L^2(X_T)_{\mathrm{gen}}$, and for $f\in C_c^\infty(\overline{T(\mathbb{A})^1})_-$ put $R(f)=\int_{\overline{T(\mathbb{A})^1}} f(y)R(y)\,dy$ and

$$\bar\chi(f)=\int_{\overline{T(\mathbb{A})^1}} f(y)\bar\chi(y)\,dy,\qquad \bar\chi\in\Pi(\overline{T(\mathbb{A})^1}).$$

The integrand $f\bar\chi$ is invariant under $\mu_2$, so $\bar\chi(f)$ is an integral over $T(\mathbb{A})^1$; if $\bar\chi$ were trivial on $\mu_2$ instead, the integral would vanish.

**Theorem 3.4.** *Let $f\in C_c^\infty(\overline{T(\mathbb{A})^1})_-$.*

*(a) $R(f)$ is the integral operator on $L^2(X_T)_{\mathrm{gen}}$ with kernel*

$$K_f(x,y)=\sum_{\delta\in T(\mathbb{Q})} f(x^{-1}\delta y),$$

*and $R(f)\bar\chi=\bar\chi(f)\,\bar\chi$ for $\bar\chi\in\Pi_{\mathrm{aut}}(X_T)$.*

*(b) $R(f)$ is of trace class, and*

$$\sum_{\bar\chi\in\Pi_{\mathrm{aut}}(X_T)}\bar\chi(f)=\operatorname{tr}R(f)=\int_{X_T}K_f(x,x)\,dx=\operatorname{vol}\big(T(\mathbb{Q})\backslash T(\mathbb{A})^1\big)\sum_{\delta\in T(\mathbb{Q})}f(\delta),\tag{3.3}$$

*the sum on the left converging absolutely.*

*Proof.* (a) For $\varphi\in L^2(X_T)_{\mathrm{gen}}$,

$$(R(f)\varphi)(x)=\int_{\overline{T(\mathbb{A})^1}}f(y)\varphi(xy)\,dy=\int_{\overline{T(\mathbb{A})^1}}f(x^{-1}y)\varphi(y)\,dy=\int_{X_T}\sum_{\delta\in T(\mathbb{Q})}f(x^{-1}\delta y)\varphi(y)\,dy,$$

because $\varphi(\delta y)=\varphi(y)$. For $\varphi=\bar\chi$ the first expression is $\bar\chi(x)\bar\chi(f)$.

(b) The group is abelian, so $f(x^{-1}\delta x)=f(\delta)$, and $\int_{X_T}K_f(x,x)\,dx=\operatorname{vol}(X_T)\sum_\delta f(\delta)$ with $\operatorname{vol}(X_T)=\operatorname{vol}(T(\mathbb{Q})\backslash T(\mathbb{A})^1)$ since $\mu_2$ has volume 1. The function $F(x)=\sum_\delta f(\delta x)$ is smooth on $X_T$ (the sum is locally finite, $T(\mathbb{Q})$ being discrete), so its Fourier series converges absolutely; its Fourier coefficient at $\bar\chi$ is $\int_{X_T}F\bar\chi=\bar\chi(f)$, which vanishes unless $\bar\chi$ is genuine. Evaluating the Fourier series at $x=1$ gives the outer equality of (3.3), i.e. Poisson summation. Since $R(f)$ is diagonal in the orthonormal basis $\operatorname{vol}(X_T)^{-1/2}\bar\chi$ with summable eigenvalues $\bar\chi(f)$, it is of trace class with that trace. □

Theorem 3.4 replaces Theorems 3.8 and 3.9 of the 1995 notes. For $n = 2$ every conjugacy class in $\overline{T(\mathbb{A})^1}$ is a point and every centralizer is the whole group, so the orbital integrals there reduce to the values $f(\delta)$. Here $f(\delta)$ means $f$ at the image $(\delta, \kappa_{\mathbb{A}}(\delta))$ of $\delta$; for $T = A$ it is $f(a(\delta), 1)$.

The trace formula for $\overline{T(\mathbb{A})^1}$ is literally Poisson summation on $T(\mathbb{Q})\backslash T(\mathbb{A})^1$, after multiplying by a genuine automorphic character:

**Lemma 3.5.** *Fix $\bar\chi_0 \in \Pi_{\mathrm{aut}}(X_T)$ with local components $\bar\chi_{0,v}$, as in Lemma 3.1(c). For an anti-genuine $\bar f$ on $\overline{T(\mathbb{A})^1}$ (resp. $\overline{T(k_v)}$) define $f$ on $T(\mathbb{A})^1$ (resp. $T_k(k_v)$) by $f(t) = \bar f(\tilde t)\bar\chi_0(\tilde t)$ (resp. $\bar\chi_{0,v}$), $\tilde t$ any lift of $t$.*

*(a) $\bar f \mapsto f$ is well defined, bijective from $C_c^\infty(\overline{T(\cdot)})_-$ onto $C_c^\infty(T(\cdot))$, and carries convolution to convolution.*

*(b) At every $v$ where $\bar\chi_{0,v}$ is trivial on $T(\mathcal{O}_v)^*$ (almost all $v$), it carries $e_{T,v}$ to the characteristic function of $T_k(k_v) \cap K_v$, which is the unit of the Hecke algebra of $T_k(k_v)$ when $T_k(k_v) \cap K_v$ has volume 1.*

*(c) $f(\delta) = \bar f(\delta)$ for $\delta \in T(\mathbb{Q})$, and $(\bar\chi_0 \cdot (\chi \circ p))(\bar f) = \hat f(\chi) := \int_{T(\mathbb{A})^1} f(t)\chi(t)\,dt$ for every character $\chi$ of $T(\mathbb{A})^1$.*

*Consequently (3.3) for $\bar f$ is the Poisson summation formula $\sum_\chi \hat f(\chi) = \mathrm{vol}(T(\mathbb{Q})\backslash T(\mathbb{A})^1) \sum_{\delta \in T(\mathbb{Q})} f(\delta)$ for $f$, $\chi$ running over the characters of $T(\mathbb{Q})\backslash T(\mathbb{A})^1$.*

*Proof.* (a) $\bar f\bar\chi_0$ is $\mu_2$-invariant because $\bar f$ is anti-genuine and $\bar\chi_0$ genuine, and the inverse map is $f \mapsto (f \circ p)\bar\chi_0^{-1}$. For convolution, $(\bar f_1 * \bar f_2)(x)\bar\chi_0(x) = \int \bar f_1(y)\bar\chi_0(y)\,\bar f_2(y^{-1}x)\bar\chi_0(y^{-1}x)\,dy$, and the integrand is $\mu_2$-invariant in $y$, so the integral over the cover equals the integral over $T$. (b) is immediate. (c) $\bar\chi_0$ is trivial on $T(\mathbb{Q})$; the second identity is the definition, the integrand being $\mu_2$-invariant. □

This replaces Lemmas 3.11 and 3.12 and the argument through Claims 1 and 2 of the 1995 notes. The local maps $\bar f_v \mapsto f_v$ involve no sum over $T(\mathbb{Q}_p)_\delta\backslash T(\mathbb{Q}_p)$ and no factor $s_{\mathbb{A}}(\delta)$, because there is no conjugation. Together with Theorem 3.4 they show that the correspondence $\chi \leftrightarrow \bar\chi_0 \cdot (\chi \circ p)$ of Theorem 3.2 matches the two trace formulas term by term.

### 3.3 Distributions for Section 6

Let $S$ be a finite set of places of $\mathbb{Q}$ and let $\overline{A(\mathbb{A}_S)}$ be the cover of $A(\mathbb{A}_S) = \prod_{p\in S} A(\mathbb{Q}_p)$ defined by $\prod_{p\in S}\beta_p$. Let $\mathcal{H}(\overline{A(\mathbb{A}_S)})$ be the algebra of functions $\bigotimes_{p\in S} f_p$, $f_p \in C_c^\infty(\overline{A(\mathbb{Q}_p)})_-$, and the analogous algebra $\mathcal{H}(\overline{A(\mathbb{A})})$ of Section 3.1. For a class $\mathcal{O} = \{\delta\}$ of $A(\mathbb{Q})$ (a single element, since the cover is abelian) and $\bar\chi \in \Pi_u(\overline{A(\mathbb{A}_S)})$, put

$$I^A_{\mathcal{O}}(f) = f(a(\delta), 1), \qquad I^A_{\bar\chi}(f) = \bar\chi(f) = \int_{\overline{A(\mathbb{A}_S)}} f(y)\bar\chi(y)\,dy, \tag{3.4}$$

for $f \in \mathcal{H}(\overline{A(\mathbb{A}_S)})$, and likewise over $\mathbb{A}$. Here $(a(\delta), 1)$ is the image of $\delta$ under the rational splitting (Lemma 2.6(b)). Both are invariant distributions: the cover is abelian, so conjugation acts trivially.

*Remark* 2026.6. For general $n$ the cover of $A$ is not abelian: by the computation in Lemma 2.2(f), $\overline{A(F)}$ is a Heisenberg-type group whose commutator pairing is $(t_1, t_2)_n^{-2}$. Its genuine irreducible representations are then parametrized by genuine characters of its center, the preimage of $A^N$ with $N = n/\gcd(n, 2)$. Matching them with characters of $A^N$ requires a splitting over $A^N$, and the global argument requires a Hasse principle for $N$-th powers, which holds when $\mu_N \subset k$. These are referee items II.1–II.4; none of them arises for $n = 2$, and this section does not treat $n > 2$.

*Remark* 2026.7. The script `checks/sec3_n2_check.py` verifies: that the 1995 cocycle on $A$ equals (3.1); that $\gamma_v$, computed from (3.2), is an eighth root of unity, depends only on the square class, is 1 on units at $p = 3, 5, 7$, and satisfies Lemma 3.3(a); that $\chi_{\psi,v}$ respects the group law (Lemma 3.3(c)) at $v = \infty, 2, 3, 5, 7$; the product formula of Lemma 3.3(b) for random $a \in \mathbb{Q}^\times$; the odd-$p$ values of Example 3.B; and Example 3.A. The rest of the section is proved in the text.

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